An ExportOriented Agro Value Optimization System for HighProfit Crop Recommendation

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***Abstract*—Agriculture is a very significant factor in a country like India, where many people rely on this sector for their daily earnings. Still, there are many problems faced by farmers, such as lower profits, changing weather, and fluctuating prices in the market. Farmers generally select their crops according to their past experiences, which may not always provide the best outcome in current scenarios. Machine learning has been implemented in recent times to help farmers in choosing their crops according to their soil conditions and climate, which will provide a good yield in their future harvests. Still, most of these machine learning algorithms are focused on crop growth, while other significant factors like prices and export are ignored, which directly affects their profit. In this paper, we will discuss a smart decision support system for farmers using machine learning and market analysis techniques, where soil nutrients like nitrogen, phosphorus, and potassium, along with weather conditions like temperature and rainfall, are taken into consideration to provide a list of suitable crops, along with expected yield and income based on local and export market prices. One of the main features of this system is that it can help farmers make a decision regarding whether they should sell their crops in the local market or export them in order to gain more profit. It can also offer farmers some special high-value crops that have a good demand in international markets. From these results, we can see that the system can offer more useful suggestions compared to traditional methods. It can help farmers make better decisions based on their knowledge and increase their profit.**

***Index Terms*—Crop Recommendation, Machine Learning, Agriculture, Export Optimization, Soil Analysis, Climate Data, Smart Farming**

I. Introduction

Agriculture is one of the main sectors in various countries like India, as it can provide food, employment, and income for people. A significant number of people depend on agriculture as a source of livelihood. Though agriculture is a significant sector in these countries, farmers face a number of problems in their daily activities. Most farmers choose a crop based on their own experience and what is grown by people in their area. Though this is a traditional way of selecting crops, it does not take into account some significant factors like soil conditions, climate changes, etc. As a result, farmers may grow crops that may not bring a good profit. With the advancement in technology, machine learning is also used to help farmers make the best decisions. Many such systems have been developed that help farmers choose the best crops according to the soil and weather conditions. In the same way, yield prediction helps farmers estimate the amount of crop that is going to be produced. But there is one major drawback in the systems that are used today. All the systems focus on the yield, but none of them consider the market-related aspects such as price, demand, and export value. In real life, the profit is not only related to the yield but also related to the place where the crop is sold. In order to overcome this drawback, we propose a system that not only helps farmers choose the best crops but also helps them make the best decisions by considering the market-related aspects. The system is designed in such a way that it follows certain steps. First, the system suggests the best crops according to the soil and weather conditions. Then, the system estimates the yield by using the yield prediction system. After that, the system compares the profit with the help of the market price. Finally, the system suggests the best place where the crop is to be sold. The system also suggests the best crops that are more valuable in the market. The crops that are suggested by the system are the ones that help the farmer earn more compared to the normal crops. Thus, the main purpose behind the design of the system is to provide the best support to the farmers so that they can make the best decisions.

II. Literature Review

In this section, a review of literature is provided for the existing research work carried out in relation to crop recommendation, yield estimation, and decision-making tools in agriculture.

A. Crop Recommendation Systems

Crop recommendation have been researched for several years in the domain of agriculture. Initially, crop recommendation systems were simple rule-based systems that used factors such as soil type, rainfall, and season. Although these systems were simple to use, they were not very accurate and could not handle complex situations. Later on, machine learning algorithms such as Decision Trees, SVM, Random Forest, etc. were used for crop prediction.

These algorithms gave better results because they can learn from data. Recently, some researchers have also used advanced algorithms such as ensemble algorithms for crop prediction.

However, most of these crop recommendation systems only take into consideration factors such as soil type and weather conditions. Market-based factors have not been considered in most of these systems.

B. Yield Prediction Models

Yield prediction is also very important for the farmer to know how much crop they can harvest. Various models, such as Linear Regression, Random Forest, Gradient Boosting, etc. have been used for this purpose. These models help in planning and managing resources in a better way. Sometimes, satellite images and time-based data are also used for prediction. But again, these systems are mainly concerned with the production aspect rather than the profit aspect.

C. Market and Price Analysis

Some research work has also been done on predicting the prices of crops with the help of past data and trends. These systems help in understanding the price trends that might occur in the future. But again, these systems are generally standalone systems, not connected with crop recommendation systems or yield prediction systems.

Therefore, the farmers are not provided with the required information.

D. Integrated Systems

Some recent research work on crop recommendation systems includes the integration of various features such as crop recommendation, yield prediction, and monitoring. These systems help in the efficient farming process.

But again, the export aspect is not considered, nor is the profit aspect compared with the local market and the international market.

E. Research Gap

Read many papers and tried apps: From the above study on existing research work, we found the following gaps:

- The existing systems are mainly concerned with crop prediction.
- Market aspects are not considered.
- Export aspects are not considered.
- Profit aspects are not considered.

To overcome the above problems, we propose a system that combines machine learning with market analysis. This system not only helps the

F. Proposed System Overview

To overcome the limitations of existing systems, we propose a smart agriculture decision support system that includes crop recommendations, yield prediction, and market analysis. The basic idea behind this system is to not only recommend a crop to a farmer, but also to maximize his profit.

The system will accept various types of inputs, such as soil nutrients, weather conditions, and location information. Based on these inputs, the system will recommend a crop to a farmer using a classification model.

Once a crop has been recommended, the system will then predict the yield using a regression model. This will help a farmer know how much he/she will be able to produce.

Finally, the system will perform a market analysis. It will determine the cost of selling the crop locally and also in foreign markets. This will also determine if a farmer should sell his/her crop locally or export it.

This will be a more useful system compared to other systems because of its agricultural and economic analysis.

G. Special Crop Recommendation

The system will not only recommend a crop to a farmer, but also recommend a special crop. Special crops refer to those crops that a farmer has not typically planted, but they have a higher market value compared to other crops.

Examples of such crops are moringa, makhana, and psyllium husk. Such crops have more profit potential compared to other crops because of the export value of these crops.

The system can identify these crops depending on factors such as market demand, scarcity, and price difference between the local market and export market. This feature of the system can help farmers explore new business opportunities and increase their income.

H. Advantages of Proposed System

The proposed system has several advantages over the traditional approach:

- It can provide both agricultural and economic information
- It can help the farmers select crops depending on the profit rather than production
- It can help the farmers in export-based decisions
- It can reduce the use of guesswork and experience
- It can help in better decision-making

The combination of these features can help the system in become more practical for use.

I. Real-World Applicability

The system is also simple and easy to use, ensuring that farmers can use the system without facing any difficulties. The basic purpose of developing such a system is to ensure that even a farmer who has basic knowledge of technology can use the system comfortably.

There is a web interface provided where a farmer can input his/her information such as soil nutrients, rainfall, temperature, location, and season. After inputting all the information, the system will process the information and provide results within a matter of seconds.

The interface is also developed in such a manner that all the steps are simple and easy to follow. There is no need for a farmer to have knowledge of complex terms, as the system will provide simple information. For example, instead of showing only results in terms of numbers, the system will also indicate what crop can be planted, what yield can be obtained, and whether to sell locally or export.

Additionally, the system can be operated using different devices such as mobile phones, tablets, or computers, making it flexible for people living in rural areas. This is because, nowadays, most farmers possess smartphones, making it more feasible for real-world application.

The system can also be extended in the future to include real-time data. For example, APIs can be integrated to include real-time weather updates such as rainfall, temperature, and climate conditions. Additionally, real-time data on current market trends, both locally and internationally, can be included. Furthermore, government schemes, subsidies, and support can also be included to assist farmers.

III. METHODOLOGY

A. Overview

The proposed system, named the Export-Oriented Agro Value Optimization System, is designed to combine machine learning models with market analysis, thereby providing an intelligent system for decision-making. The system is designed to cover both agronomic and economic aspects with the aim of selecting the best crop for maximum profit. The system architecture is composed of the following components: - Crop prediction model for crop selection - Yield prediction model for yield estimation - Market analysis model for price estimation - Profit optimization model for intelligent decision-making - Special crop recommendation module for export purposes

Given an input feature vector:

$$x \in \{R\}^n \tag{1}$$

the system learns the mapping:

$$f(x) \to (c, y, P_l, P_e, Pi_l, Pi_e, D) \tag{2}$$

Where:

- c = predicted crop
- y = predicted yield
- Pl,Pe= local and export prices
- Pi l, Pi e= profits
- D = final decision

B. Dataset

The proposed system utilizes different datasets, which are mentioned below:

- Crop recommendation dataset (soil, climate, etc.)
- Yield dataset (crop production statistics)
- Market price dataset (local prices)
- Mapped export dataset (export prices, countries, etc.)

The dataset consists of different features, i.e., soil nutrients, temperature, rainfall, humidity, etc.

To avoid bias, the following techniques will be used:

- ing values will be handled using different techniques
- Normalization will be performed on the dataset
- The dataset will be divided into training and testing sets (80:20)

C. Feature Representation

The feature vector input is described as follows:

$$x = (N, P, K, T, R, H, L) \tag{3}$$

where:

- N, P, K = soil nutrients
- T = temperature
- R = rainfall
- H = humidity
- L = location

All these are important in defining the environment in which the crops will grow.

D. Crop Prediction Model

The crop prediction problem is viewed as a classification issue. A Random Forest classifier is utilized because of its ability to handle non-linear relationships and achieve high performance.

The crop prediction is defined as:

$$c = \arg\max_{y_i} P(y_i \mid x) \tag{4}$$

In the above equation, the model predicts the crop with the highest probability.

In the proposed model, a combination of multiple decision trees(Random Forest) is utilized to improve generalization by reducing overfitting.

1) Random Forest for Crop Prediction:

The algorithm Random Forest is employed in crop prediction because of its strength and accuracy. The algorithm employed in this model follows an ensemble learning approach, where decision trees are employed together to generate accurate results.

In this algorithm, a random subset of data from the data set is employed to train a decision tree. Additionally, a random subset of features is employed in every node during splitting, where data splitting takes place into two classes.

The final results are obtained through majority voting:

$$C = \text{mode}(T_1(x), T_2(x), \ldots, T_n(x)) \tag{5}$$

Where:

- Prediction of ith tree is represented by Ti(x)
- n is the number of trees
- This approach reduces variance, prevents overfitting, and improves generalization.

2) Decision Tree Splitting Criterion:

Each decision tree is constructed using a recursive partitioning method. At every node in the tree, an optimal feature is selected using an impurity measure defined using the Gini Index.

$$\text{Gini} = 1 - \sum_{i=1}^{c} p_i^2 \tag{6}$$

A low Gini Index signifies a high purity at nodes. An optimal feature that results in minimum impurity is selected to facilitate efficient classification.

E. Yield Prediction Model

The yield prediction is performed by regression techniques such as the Gradient Boost.

The yield estimation is given by the formula:

$$y = \beta_0 + \sum_{i=1}^{n} \beta_i x_i + \epsilon \tag{7}$$

where:

- y is the predicted yield,
- β_i are the model coefficients,
- ϵ is the error term.

This model represents the relationship between environmental factors and crop yield.

1) Gradient Boosting for Yield Prediction:

Gradient Boosting is used to boost the accuracy of the model by sequentially adding models.

Each new model corrects the errors of the previous model, resulting in improved accuracy.

2) Loss Function

The model is optimized using a loss function referred to as Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{8}$$

This measures the difference between actual and predicted values.

F. Economic Analysis and Decision Module

The economic analysis module is used to determine the profitability of crop production through an integration of yield prediction, market pricing, and cost calculation. This module transforms the system from a predictive model to a decision-support system.

The revenue generated through crop production is determined using the following equation:

$$R = y \times P \tag{9}$$

In this equation, y is the predicted yield per unit area and P is the market price per unit crop. As this system takes into consideration both local and export markets, revenue is determined using two different equations:

$$R_l = y \times P_l, \quad R_e = y \times P_e \tag{10}$$

In these equations, P_l and P_e represent local and export market prices per unit crop, respectively.

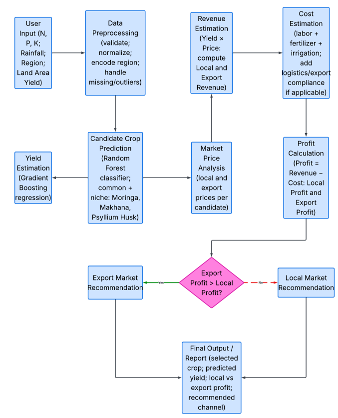


Fig. 1. Workflow of the Proposed Agro-Export Value Optimization System

The flowchart explains how the system processes input data and makes profit-based decisions for crop selection and market recommendation

To make the system more accurate in determining profit, the total cost of production is calculated using the following equation:

$$C = C_{labor} + C_{fertilizer} + C_{irrigation} \tag{11}$$

The net profit is determined using the following equation:

$$\pi = R - C \tag{12}$$

To make the system more accurate in terms of determining local and export market profits, the profit is determined using two different equations:

$$\pi_l = R_l - C, \quad \pi_e = R_e - C \tag{13}$$

The decision-making process is based on maximizing the profit using the following equation:

$$D = \begin{cases} \text{Export}, & \text{if } \pi_e > \pi_l \\ \text{Local}, & \text{otherwise} \end{cases} \tag{14}$$

In addition to this equation, an export suitability score is used to determine those crops that have high market value in other countries:

$$S = w_1 \cdot \text{Demand} + w_2 \cdot \text{Rarity} + w_3 \cdot \text{PriceDifference} \tag{15}$$

This equation ensures that the system selects crops that are suitable not only for local market conditions but also economically beneficial for export purposes.

IV. RESULTS AND DISCUSSIONS

A. Experimental Setup

The proposed model is tested on multiple agricultural datasets including soil characteristics, crop yield, and crop prices. Since proper datasets for export prices are not always available, mapped datasets for export price and export country are used. To ensure proper predictions, data pre-processing techniques such as handling missing values and balancing data, if required, are employed. The data is then utilized for model building.

The data sets are split into training data sets and testing data sets. Machine learning algorithms such as Random Forest and Gradient Boosting are employed.

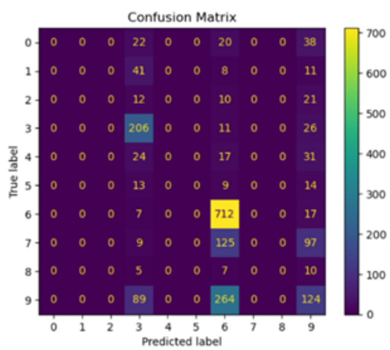
Important features such as soil nutrients, temperature, humidity, rainfall, etc., are taken into consideration. Model performance is evaluated using metrics such as accuracy, precision, recall, and F1-score.

B. Quantitative Results

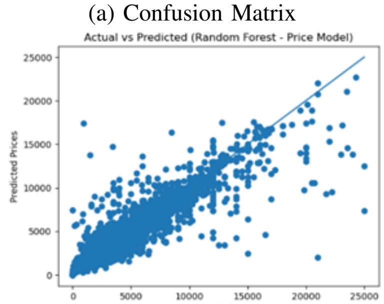
The performance of the proposed models is shown in Table I.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT MODELS

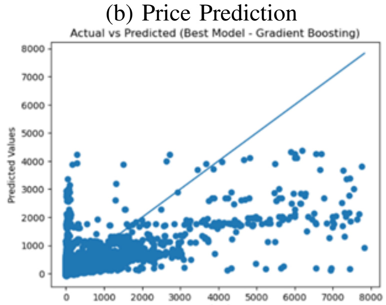
Task	Best Model	Accuracy	Precision	Recall	F1	MAE	RMSE	R^2
Yield Prediction	Gradient Boosting	-	-	-	-	228.130	587.821	0.366
Soil Classification	Random Forest	0.8085	0.6890	0.8085	0.6894	-	-	-
Price Prediction	Random Forest	-	-	-	-	627.153	1341.361	0.842



(a) Confusion Matrix



(b) Price Prediction



(c) Yield Prediction

Fig. 2. Model Performance Results

Fig. (a) shows the classification performance of the model, where higher values along the diagonal indicate correct predictions.

Fig. (b) shows a strong correlation between actual and predicted prices, indicating good accuracy.

Fig. (c) shows a moderate correlation between actual and predicted yield values, indicating reasonable performance.

C. Comparative Analysis with Existing Methods

The proposed model illustrates improved performance compared to old existing agricultural prediction models by integrating multiple models into a single framework, Unlike existing models which focus on a single task such as crop recommendation, soil classification, yield prediction, local price prediction, export price prediction and export country along with the profits

Existing models basically rely on basic machine learning algorithms and very limited preprocessing techniques so accuracy is moderate and reduced generalization. The proposed model used advance machine learning algorithms and proper preprocessing and feature selection with resulted in enhanced performance unlike traditional Systems

In the proposed model the integration of multiple models allows the system to capture complex relationships which are often overlooked and missed in existing models

TABLE II
COMPARISON WITH EXISTING METHODS

Aspect	Existing Methods	Proposed Model
Approach	Single-task models	Multi-task integrated model
Prediction Scope	Crop, yield, or export price only	Crop, soil, yield, price, export price, export country
Algorithms Used	Basic ML (Linear, SVM)	Advanced ML (Random Forest, Gradient Boosting)
Accuracy	Moderate	Improved performance
Data Handling	Limited preprocessing	Feature selection + normalization
Model Integration	Independent models	Unified end-to-end system
Practical Use	Theoretical focus	Real-world farmer support
Decision Support	Limited	Comprehensive decision-making

D. Ablation Study

An ablation study is carried out to check the performance of different machine learning models on different tasks. From the results, it can be seen that the performance of the ensemble models is better compared to the basic models. For the soil classification task, the performance of the Random Forest model is the best, with high accuracy compared to the KNN and SVM models, which have low accuracy.

For the price prediction task, the performance of the Random Forest model is the best, with the highest R2 value. This means that the model can predict prices effectively. It can be used with different types of data. For the yield prediction task, the performance of the Gradient Boosting model is the best, and it can be used with complex data.

From the results, it can be seen that the performance of the ensemble models is better compared to the basic machine learning models. These models can be used to solve different problems in agriculture.

Furthermore, the comparative evaluation also points out that simple models are not able to handle complex patterns in the data, which is a characteristic of agricultural data. The ensemble methods are also helpful in reducing variance and bias by aggregating different learners.

The improvement in ensemble methods for both classification and regression problems is an indicator of the flexibility of these methods. The results are consistent and support the use of ensemble methods for practical problems that are prone to variability and noise.

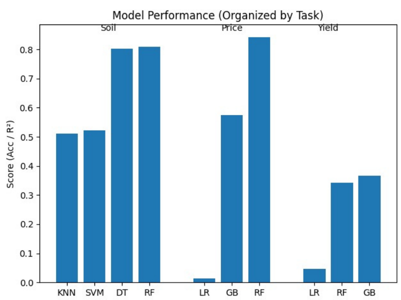


Fig. 3. Summary of Best Performing Models Across Tasks

E. Cross-task Evaluation

Cross-task experiments are conducted to evaluate the ability of the proposed model to generalize across different agricultural tasks. The results are summarized in Table III.

TABLE III
CROSS-TASK EVALUATION RESULTS

Task	Model Used	Performance
Yield Prediction	Gradient Boosting	$R^2 = 0.3662$
Soil Classification	Random Forest	Accuracy = 0.8085
Price Prediction	Random Forest	$R^2 = 0.8429$

The results show that the proposed model maintains consistent performance across different tasks and data conditions. In price prediction, the model achieved a high R^2 score of 0.8429, indicating strong predictive capability. The soil classification model achieved an accuracy of 0.8085, demonstrating reliable classification performance.

Although the yield prediction model has a comparatively lower R^2 score, it still effectively captures complex relationships in agricultural data, highlighting the robustness of the proposed system.

F. Discussion

The findings of the experiment highlight several important observations. First, ensemble-based models perform better than traditional models across multiple tasks, as they can capture complex nonlinear relationships in the data.

Second, different models perform best for different tasks. There is no single model that performs best in all cases. In the proposed system, Gradient Boosting performs best for yield prediction, while Random Forest performs well for both soil classification and price prediction.

Third, integrating multiple models into a single system improves decision-making capability compared to existing approaches.

Overall, the proposed model demonstrates robustness, scalability, and real-world applicability, making it a valuable system for helping farmers make data-driven decisions.

V. CONCLUSION

This paper proposes a decision support system in agriculture based on various machine learning models. The proposed system uses ensemble techniques like Random Forest and Gradient Boosting, which work efficiently in handling non-linear relationships between data.

The results obtained from experiments show that the proposed model works better than existing models. Random Forest achieves an accuracy of 0.8085 in soil classification and an R^2 score of 0.8429 in price prediction. Gradient Boosting performs best in yield prediction with an R^2 score of 0.3662.

The ablation study highlights the importance of selecting appropriate models for different tasks. Ensemble methods consistently outperform traditional algorithms by providing better accuracy and generalization across multiple tasks.

Overall, the proposed system provides a balance between performance, simplicity, and practical usability. By integrating multiple models into a single framework, it offers an efficient solution for supporting farmers in decision-making.

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